

## Type I Diabetes & Insulin

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Block: Endocrine/Urology/Reproductive

Thread: Pharmacology

Conflict of Interest: Dr. Simon has no Conflicts of interest to disclose

### Learning Objectives

By exam time, you should be able to do the following:

1. Distinguish the essential pathophysiological differences between type 1 and type 2 diabetes and contrast the role of insulin treatment in these different types of diabetes
2. Explain the pattern of endogenous insulin secretion from the islet cells and how to prescribe exogenously administered insulin to replicate this pattern (basal -bolus concept)
3. Identify the insulin types (basal vs. mealtime) and differentiate their pharmacokinetic profiles
4. Appraise and contrast the differences between human insulins and insulin analogs
5. Describe adverse effects associated with insulin therapy

### Further Reading

1. [Simmons KM. Type 1 diabetes : A predictable disease. \*World journal of diabetes.\* 2015;6\(3\):380-390. doi:10.4239/wjd.v6.i3.380](#)
2. [Hirsch IB, Juneja R, Beals JM, Antalis CJ, Wright EE. The Evolution of Insulin and How it Informs Therapy and Treatment Choices. \*Endocrine reviews.\* 2020;41\(5\):733-755. doi:10.1210/endrev/bnaa015](#)

## Study Questions

1. A 28-year-old man presents with polyuria, polydipsia, blurred vision, and a 10-pound weight loss, all of recent onset over the past few weeks. He has no past medical history. He is not taking any medications. On exam, his BMI is 21 kg/m<sup>2</sup>, blood pressure 110/60 mm Hg, pulse 98 bpm, his skin is dry; otherwise unremarkable. You suspect diabetes mellitus.
  - What test(s) should you order to confirm the diagnosis?
  - What type of diabetes does this patient have? How could you confirm it?
  - What treatment would you start?
  
2. A 35-year-old woman has type 1 diabetes and is on basal and mealtime insulins. Which of the following insulins would she take once a day as a basal insulin?
  - Aspart
  - Degludec
  - Glulisine
  - Regular
  - Lispro
  
3. A 24-year-old with Type 1 diabetes since childhood asks you to change their analog basal insulin glargine to human insulin NPH due to cost concerns. Which statement is correct about switching to NPH insulin?
  - It may increase the risk of hypoglycemia
  - It may lead to weight loss
  - NPH insulin should be dosed once daily
  - NPH insulin should be dosed just prior to each meal

## A. Definition and Classifications of Diabetes

Diabetes Mellitus is a chronic metabolic disorder characterized by hyperglycemia that results from a variety of pathophysiologic mechanisms

### 1. Type 1 Diabetes

- a. Autoimmune disorder
- b. Pancreatic  $\beta$ -cell destruction with essentially complete insulin deficiency
- c. Typically presents as acute onset with symptoms
- d. 5% of all diabetics. Higher incidence in Caucasians
- e. More commonly in young, but can occur at any age

### 2. Type 2 Diabetes

- a. A complex metabolic disturbance
- b. Insulin Resistance affecting glucose and lipid metabolism combined with a failure of adequate  $\beta$ -cell insulin secretion
- c. Develops insidiously over many years
- d. Frequently asymptomatic
- e. Intermediate diagnoses – “Pre-Diabetes”
  - i. Impaired glucose tolerance (IGT)
  - ii. Impaired fasting glucose (IFG)

### 3. Gestational Diabetes

- a. Diabetes that develops during pregnancy and usually subsides after delivery
- b. Pathogenesis - In the setting of insulin resistance characteristic of pregnancy, diabetes can occur if beta cell insulin secretion is inadequate
- c. Incidence of up to 6% of pregnancies
- d. If untreated, can lead to increased fetal morbidity and mortality
- e. 30-40% will develop type 2 diabetes 5-10 years later
- f. Women at risk should be screened between weeks 24 and 28

### 4. Other Forms of Diabetes

- a. Pancreatic disease (hemochromatosis, pancreatitis) – can result in destruction of pancreatic islet tissue
- b. Endocrinopathies - due to excess secretion of hormones that antagonize the action of insulin: cortisol (Cushing's), growth hormone (acromegaly), catecholamines (pheochromocytoma), glucagon (glucagonoma)
- c. Drug therapy (example: corticosteroids increase insulin resistance)

### 5. Diabetes due to Defined Genetic Mutations

- a. Rarely, mild forms of non-immune mediated diabetes can be associated with a specific genetic defect in a protein known to be important for insulin secretion or insulin action
- b. A typical example is a rare genetic defect affecting glucose-stimulated insulin secretion, called MODY or “Maturity Onset Diabetes of Youth”

## B. Acute Symptoms of Diabetes

1. Failure to utilize calories in diet because of a lack of anabolic effect of insulin. Hyperglycemic symptoms include:
  - a. Polyuria (increased urination)
  - b. Polydipsia (thirst)
  - c. Dehydration
  - d. Polyphagia (hunger)
  - e. Weight loss
  - f. Blurred vision
  - g. Altered mental status (hyperosmolar states)

## C. Criteria for Diagnosis of Diabetes

1. FPG (fasting plasma glucose)  $\geq 126$  mg/dl. Fasting is defined as no caloric intake for at least 8 h.
2. 2-hour plasma glucose  $\geq 200$  mg/dl during an OGTT (oral glucose tolerance test using 75-grams of glucose)
3. In a patient with classic symptoms of hyperglycemia or hyperglycemic crisis, a random plasma glucose  $\geq 200$  mg/dl
4. HbA1c  $\geq 6.5\%$

Criteria A, B and D should be confirmed by repeat testing

- a. Oral glucose tolerance test (OGTT) criteria (75 gm oral glucose given fasting)
  - i. 140-199 mg/dL at 2 hr = Impaired Glucose Tolerance, a form of “Pre-Diabetes”
  - ii. Glucose  $< 140$  mg/dL at 2 hr = Normal Glucose Tolerance
- b. Diagnosis by Hemoglobin A1c (3 month “average” of glucose control)

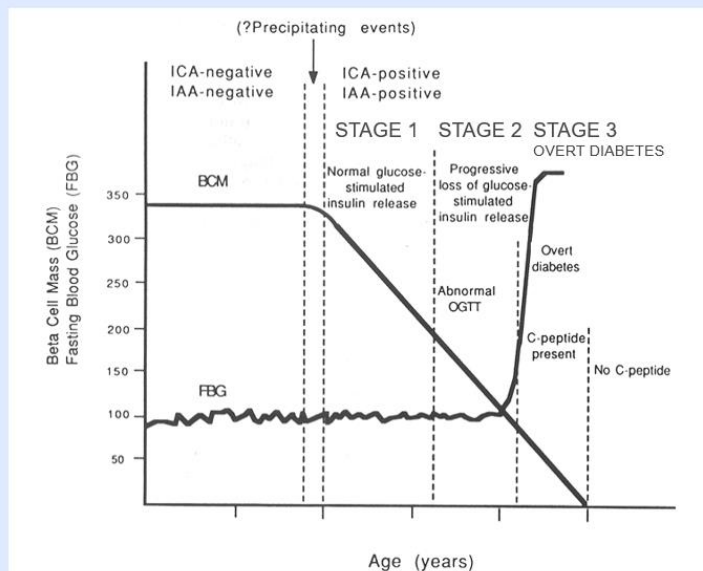
- i. Non-enzymatic glycation of hemoglobin; the degree of glycation reflects the average glucose exposure over the half-life of the RBC. A1C is therefore influenced by red cell survival
  - 1. Normal: < 5.7%
  - 2. Increased risk of diabetes: 5.7-6.4% "Pre-Diabetes"
- ii. A1c is also used to monitor control of diabetes- Should generally be 7.0% or less (American Diabetes Association)

## D. Type 1 Diabetes Pathogenesis

Autoimmune destruction of beta cells, set off by an environmental trigger in a genetically susceptible person

1. Genetic Component
  - a. Strong HLA association (HLA-DR 3 and 4 in 90-95% of patients), multiple genes may be involved (polygenic inheritance)
2. Environmental component
  - a. A trigger that sets off immune cascade (? virus, unknown)
3. Autoimmune (T cell mediated) destruction of  $\beta$ - cells. defects in immune recognition of  $\beta$ -cell antigens
  - a. Insulinitis: lymphocytic inflammatory reaction in islets
  - b. Antibodies to islet cell components are markers of immune destruction measurable in up to 90% of patients (can identify in laboratory testing)
    - i. GAD65- Glutamic Acid Decarboxylase 65, IA2- tyrosine phosphatases, islet cell antibodies, Zinc transporter 8
  - c. Prone to other autoimmune diseases (thyroid, Addisons, vitiligo)
4. Eventually, > 90% of  $\beta$ -cells are destroyed (hyperglycemia develops with possible ketoacidosis)

## Pathogenesis of Type I Diabetes



The pathophysiologic development of type 1 diabetes follows a well-defined pattern, as shown in this figure (adapted from initial description of the pathogenesis in 1986 by Eisenbarth et al. *New Engl. J. Med.* 314:1360, 1986).

5. Hypothetical scheme for the development of type 1 diabetes (above). The disease develops from an initial genetic susceptibility and ends with complete  $\beta$ -cell destruction. The presence of early immunological markers before the onset of  $\beta$ -cell failure has provided a basis for screening and possibly preventing Type 1 diabetes. ICA = islet cell antibodies; IAA = insulin autoantibodies

## E. Predictability of Type 1 Diabetes

1. Risk of developing type 1 diabetes:
  - a. In the general population = 0.3 - 0.5%
    - i. With one first degree relative = 2-6%
  - b. In monozygotic twins = 30-40%;
    - ii. up to 70% in twins with HLA haplotypes DR3 / DR4
  - c. If mother has type 1 diabetes = 2-3%
  - d. If father is affected = 5-6%

2. Prevention of Type 1 Diabetes
  - a. Large trials have tested whether progression to type 1 diabetes could be prevented
    - i. Individuals at high risk
    - ii. Usually first-degree relatives with (+) antibodies
  - b. Randomized to various agents
    - i. Low dose insulin injections - ? Tolerizing effect to insulin antigen
    - ii. Immune modulators
  - c. Teplizumab (Tziel)
    - i. First agent approved (in 2022) to **DELAY (not prevent)** the onset of Type 1 by up to 2 years
    - ii. Mechanism of action: binds to CD3 on T cells, to prevent signaling that leads immune destruction of the beta cells
    - iii. Indications: stage 2 diabetes- (at least 2 antibodies and dysglycemia) not overt diabetes
    - iv. Infusion daily over 14 days, High cost
    - v. Side effects- include lymphopenia (monitor WBC during treatment), infection

**Summary of Differences between Type 1 and Type 2 Diabetes**

|                                               | <b>TYPE 1</b>                            | <b>TYPE 2</b>                                        |
|-----------------------------------------------|------------------------------------------|------------------------------------------------------|
| Out-Dated names                               | “Insulin-Dependent”<br>Juvenile Diabetes | “Non-Insulin Dependent”<br>Adult-Onset               |
| Out-Dated acronyms                            | IDDM                                     | NIDDM                                                |
| Age of onset (most often)                     | Usually younger                          | Usually adults                                       |
| Ketosis                                       | Present                                  | Absent                                               |
| Recent Weight Loss                            | Yes                                      | Usually no                                           |
| Time course of onset                          | Abrupt with symptoms                     | Prolonged (over years)<br>typically with no symptoms |
| Overweight/obese                              | Not usually                              | Most of the time                                     |
|                                               | <b>TYPE 1</b>                            | <b>TYPE 2</b>                                        |
| Insulin Resistance                            | No                                       | Yes                                                  |
| Islet Cell Antibodies &<br>GAD-65 Antibodies  | Yes                                      | No                                                   |
| Insulin Secretion<br>Defect                   | Yes                                      | Yes                                                  |
| HLA-Associated<br>Antigens                    | Yes                                      | No                                                   |
| Associated with other<br>Auto-immune Diseases | Yes                                      | No                                                   |
| Insulin Rx Necessary                          | Absolutely                               | Sometimes                                            |
|                                               |                                          |                                                      |

## **F. Insulin Therapy**

### **1. Who Needs Insulin Therapy?**

- a. Type 1 Diabetes - ALL patients
  - i. Insulin always needed to prevent ketoacidosis
- b. Type 2 Diabetes
  - i. Hyperglycemia despite maximally effective doses of oral or non-insulin injectable agents
  - ii. Significant hyperglycemia at presentation
  - iii. Acute sickness, Hospital stay, Surgery
  - iv. Pregnancy

### **2. Clinical Use of Insulin**

- a. Advantages
  - i. Can control all patients
  - ii. Flexibility in dosing
  - iii. Analog long and rapid-acting insulins provide clinical benefit
- b. Disadvantages
  - i. Injections required
  - ii. Hypoglycemia
  - iii. Weight gain
  - iv. Cost

### **3. Endogenous Insulin Normal physiology**

- a. Continuous release of insulin throughout the day ('basal insulin')
  - i. Regulates lipolysis and the output of hepatic glucose
- b. Spikes of insulin release in response to a meal ('boluses')
  - i. Occurs in a robust first-phase secretion and then a more prolonged second-phase release into the portal circulation

### **4. Types of Insulin**

- a. 1921: Discovery of Insulin. Animal insulins: porcine, bovine
- b. 1980's: Purification of animal insulin, recombinant Human insulins:
  - Regular
  - NPH

- c. 1990's: Analog insulin developed: Human insulin changed by recombinant DNA techniques -reversing or substituting amino acids in human insulin:

Lispro (Humalog®, Lyumjev®)

Glargine (Lantus®)

Aspart (Novolog®, Fiasp®)

Degludec (Tresiba®)

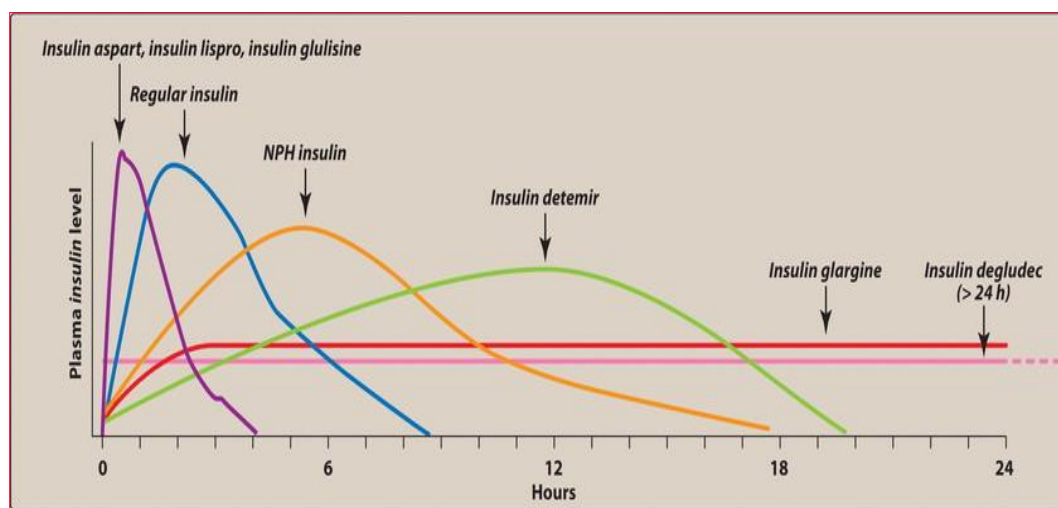
Glulisine (Apidra®)

(Note Detemir (Levemir®) discontinued in 2024)

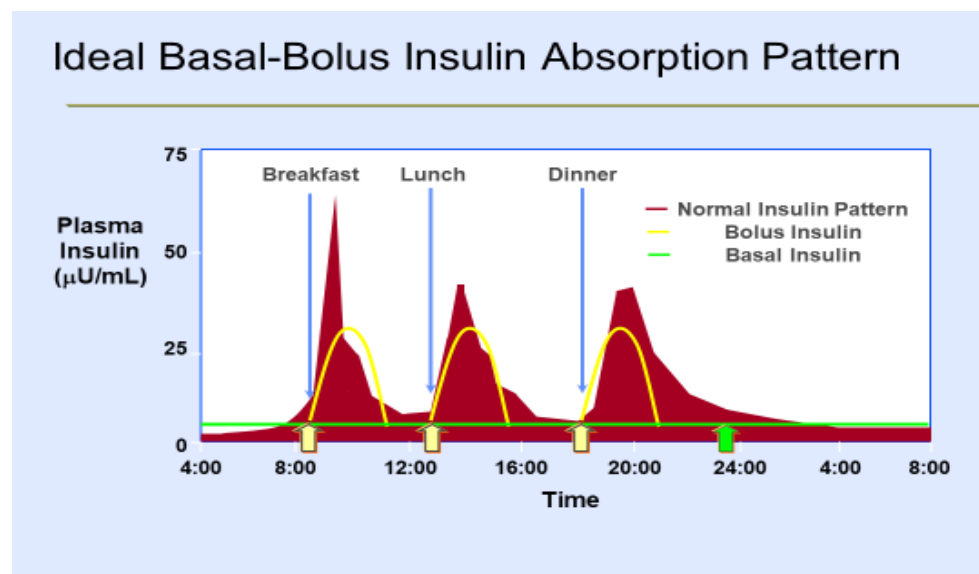
## Human & Analog Insulins

| Type                  | Name                             | Onset     | Peak         | Duration |
|-----------------------|----------------------------------|-----------|--------------|----------|
| Rapid-acting analogs  | lispro, aspart, glulisine        | 15 min    | 1-2 h        | 3-5 h    |
| Short-acting          | Regular                          | 30-60 min | 2-4 h        | 6-8 h    |
| Intermediate-acting   | NPH (neutral protamine Hagedorn) | 2-4 h     | 4-10 h       | 12-16 h  |
| Basal insulin analogs | glargine                         | 2-3 h     | ("peakless") | 20-24 h  |
|                       | degludec                         | 2-3 h     |              | 14-18 h  |
|                       |                                  | 1 h       |              | ≥ 24 h   |

## Time Action Profiles of Insulin



5. **Basal-Bolus:** Using insulin injections to try to mimic the ‘normal’ pancreas- a more physiologic regimen
  - a. Also called Multiple Daily Injections (MDI)
  - b. “Basal” is intermediate or long acting insulin
    - i. Suppresses hepatic glucose output and ketogenesis
    - ii. Approximately 50% of total insulin requirement
  - c. “Bolus” is the short or rapid acting insulin prior to the meal
    - i. Faster delivery into bloodstream as food is absorbed



## 6. Insulin Regimens (by Insulin Type)

- a. Basal Insulin (Analog)
  - i. Insulin glargine (Lantus®, Basaglar®), degludec (Tresiba®)
    1. Structurally altered insulin analogs
    2. Glargine - Soluble at acid pH, forms micro-precipitates in subcutaneous space that retard absorption
    3. Degludec forms multi-hexamers when injected into the subcutaneous tissue resulting in a subcutaneous insulin depot
  - ii. Insulin icodec - abae (Awiqli) ®
    1. First FDA approved ONCE WEEKLY insulin

2. Reversibly binds to albumin c=via diacid side chain and slowly dissociates
  - iii. They are “peakless” compared to the older insulin NPH
    1. Less hypoglycemia, less weight gain
- b. Rapid-Acting Insulin (Analogues)
  - i. Insulin lispro (Humalog®), aspart (Novolog®), glulisine (Apidra®)
  - ii. Structural analogs of insulin that facilitate dissociation from Zn crystal
  - iii. Faster onset of action and shorter duration of action
    1. Limits postprandial hyperglycemic peaks
    2. Minimizes late postprandial hypoglycemia
    3. Convenient administration immediately prior to meals
  - iv. Even faster analogues- still prefer to dose pre-meal and not after
    1. Insulin lispro aabc (Lyumjev®)- prostacyclin (trepostinil) and citrate to increase local vascular permeability and vasodilation
    2. Insulin aspart (Fiasp®)- vitamin B3 (niacinamide) which causes a transient local vasodilation
- c. Intermediate Acting Insulin NPH (Human Insulin)
  - i. Older insulin that was used frequently as basal before analogs
  - ii. Crystallized human insulin with protamine and zinc to delay absorption
  - iii. Does not mimic physiology of basal profile
  - iv. Less than 24-hour duration
  - v. Pronounced peaks
  - vi. Can cause unpredictable hypoglycemia
    1. Which will limit insulin adjustments
    2. Can lead to weight gain
- d. Short Acting Insulin Regular (Human Insulin)
  - i. Older insulin that was used as bolus before analogs
  - ii. Crystalline soluble zinc insulin
  - iii. Slower onset of action
    1. Inconvenient administration: 30 to 45 minutes prior to meal
    2. Risk of hypoglycemia if meal is delayed

3. Mismatch with postprandial hyperglycemic peak
- iv. Long duration of activity- Up to 6-8 hours
  1. Potential for late postprandial hypoglycemia
- e. Notes on Insulin Use
  - i. We still use Human Insulins
    1. Cost is lower
    2. Time action profile may match other circumstances (ex. NPH may target late day hyperglycemia in patients on morning steroids)
  - ii. Basal Bolus/ MDI insulin is NOT Perfect
    1. Using fixed amounts of mealtime insulin is not as physiologic
    2. Ex. A patient on glargine 30 units daily and aspart 5 units with meals

Monday morning ate cereal and fruit (40 carbohydrates)

Tuesday morning ate eggs and bacon (0 carbohydrates)
    3. Must assess patient: compliance (4 shots per day), provide education
    4. Best way to do this is Carbohydrate counting: how many units of insulin will cover the amount of carbs being ingested
- f. Premixed Insulin Formulations
  - i. Not all patients are willing to inject multiple times daily. Premixed insulins are another option (less optimal) which may offer convenience (2 shots per day)
  - ii. Combine rapid or short-acting insulin with an intermediate-acting insulin
  - iii. For example:
    1. Human 70/30 = 70 Unit/ml NPH + 30 Unit/ml Regular
    2. Humalog Mix 75/25= 75 Unit/ml lispro-protamine complex (intermediate acting) + 25 Unit/ml lispro
    3. Humalog Mix 50/50 = 50 Unit/ml lispro-protamine complex (intermediate acting) + 50 Unit/ml lispro
    4. Novolog Mix 70/30= 70 Unit/ml aspart-protamine complex (intermediate acting) + 30 Unit/ml aspart
- g. Concentrated Insulins
  - i. Patients on large doses of insulin

- ii. Can get same amount of units, but in a lower volume
  - 1. U-500 Insulin (Humulin® R U-500)
  - 2. Glargine U-300 (Toujeo® U-300)
  - 3. Lispro U-200 (Humalog® U-200)
- h. Inhaled Insulin
  - i. Inhaled human insulin (Afrezza®): ultra-rapid acting human insulin immediate onset, peak in 15 minutes, duration of 3 hours
  - ii. Not commonly used
  - iii. Contraindications (smokers, chronic lung disease). Need to monitor pulmonary functions tests at baseline and periodically
  - iv. Most need repeated loading of cartridges to make one mealtime dose

## 7. Continuous Subcutaneous Infusion of Insulin (CSII)

- a. Insulin infusion by a programmable mechanical pump delivered through a subcutaneous catheter
- b. Advantages:
  - i. Basal insulin delivery can be carefully adjusted over a 24-hour cycle
  - ii. Easily delivered bolus doses of insulin at mealtimes and prior to snacks
- c. Disadvantages:
  - i. Cost: pumps are expensive
  - ii. Some patients do not like idea of being attached to a device

## 8. Continuous Glucose Monitors/ Monitoring

- a. A device that can measure interstitial fluid glucose (subcutaneous). Lags behind capillary glucose
- b. Can see glucose trends
- c. Important tool for managing type 1 diabetes
- d. Cost / accessibility – not available to everyone
- e. Newer automated insulin delivery systems: continuous glucose monitor information goes to insulin pump and algorithm helps with insulin delivery
  - i. Not an artificial pancreas

## Answers to Study Questions

*These answers are not meant to be all encompassing; rather they are examples of the types of thought processes needed for application of the material.*

1. A 28-year-old man presents with polyuria, polydipsia, blurred vision, and a 10-pound weight loss, all of recent onset over the past few weeks. He has no past medical history. He is not taking any medications. On exam, his BMI is 21 kg/m<sup>2</sup>, blood pressure 110/60 mm Hg, pulse 98 bpm, his skin is dry; otherwise unremarkable. You suspect diabetes mellitus.
  - What test(s) should you order to confirm the diagnosis?
  - What type of diabetes does this patient have? How could you confirm it?
  - What treatment would you start?

**ANSWER:** *This patient has a typical presentation for type 1 diabetes: acute/recent onset of classical symptoms of diabetes (polyuria, polydipsia, blurred vision, weight loss) in a young slim patient. The weight loss is due to the catabolic state induced by the lack of insulin. Since he is symptomatic, only a random plasma glucose of at least 200 mg/dL would confirm the diagnosis. If a test is needed to confirm type 1 diagnosis, then a blood test for GAD-65 antibodies can be obtained although in this case, the diagnosis of type 1 diabetes is very straightforward and clear. Management includes basal and mealtime insulin for a total of 4 injections daily.*

2. A 35-year-old woman has type 1 diabetes and is on basal and mealtime insulins. Which of the following insulins would she take once a day as a basal insulin?
  - Aspart
  - Degludec
  - Glulisine
  - Regular
  - Lispro

**ANSWER:** *From all the insulins listed, only degludec is a basal insulin, given once daily. All other insulins (aspart, glulisine, regular and lispro) are mealtime/bolus insulins given at the time of food intake.*

3. A 24-year-old with Type 1 diabetes since childhood asks you to change their analog basal insulin glargine to human insulin NPH due to cost concerns. Which statement is correct about switching to NPH insulin?
- It may increase the risk of hypoglycemia
  - It may lead to weight loss
  - NPH insulin should be dosed once daily
  - NPH insulin should be dosed just prior to each meal

**ANSWER:** *The change to NPH insulin may increase the risk for hypoglycemia. Unlike the basal insulin glargine which is relatively peakless, NPH has a pronounced peak. This may also increase the risk for weight gain due to over-insulinization compared to glargine. Type 1 diabetics need 24-hour basal coverage to prevent DKA. The duration of action of NPH is 12-16 hours, therefore it should be dosed twice daily to ensure all-day coverage. Mealtime rapid acting insulins should be dosed prior to meals.*